Tutorial 3

Q1. What are the differences of FETs and BJTs?

Q2: Explain the construction and operation of a n/p-channel JFET.

Q3. Draw the characteristics curve of a JFET and explain.

Question 1:

For the JFET circuit in Figure 1, $V_{GS(off)}$ = -4 V and I_{DSS} = 12 mA. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation when V_{GS} = 0 V.

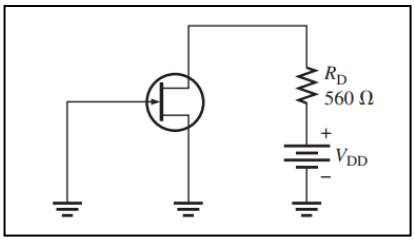


Figure 1

Question 2

a) A typical JFET has the following parameters: I_{DSS} = 9 mA and $V_{GS(off)}$ = -8 V (maximum). Using these values, determine the drain current for V_{GS} = 0 V, -1 V and -4 V.
b) A typical JFET has the following parameters: I_{DSS} = 3.0 mA, $V_{GS(off)}$ = -6 V maximum, and $g_{fs(max)}$ = 5000 μS. Using these values, determine the forward transconductance for V_{GS} = -4 V, and find I_D at this point
c) A certain JFET has an I_{GSS} of -2 nA for V_{GS} = -20 V. Determine the input resistance

Question 3:

a) Find V_{DS} and V_{GS} in the JFET circuit in Figure 2.

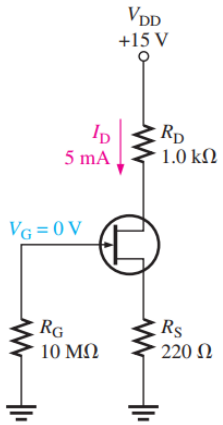


Figure 2

b) Determine I_D and V_{GS} for the JFET with voltage-divider bias in the circuit in Figure 3, given that for this particular JFET the parameter values are such that $V_D \approx 7$ V.

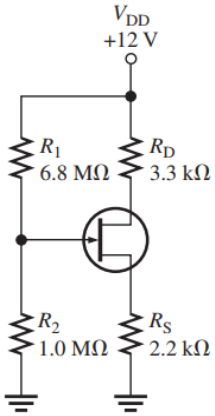


Figure 3

Question 4

a) A typical E-MOSFET gives $I_{D(on)}$ = 500 mA (minimum) at V_{GS} = 10 V and $V_{GS(th)}$ = 1 V. Determine the drain current for V_{GS} = 5 V.

b) For a certain D-MOSFET, I_{DSS} = 10 mA and $V_{GS(off)}$ = -8 V. Is this an *n*-channel or a *p*-channel D-MOSFET? Calculate I_D at V_{GS} = -3 V. Calculate I_D at V_{GS} = +3 V.

.